

AIQG — Freedom-to-Operate Claim-Read

Status: Technical FTO triage — 2026-06-19. Granted **independent** claims of four patents read against the AIQG attribution design. **Not legal advice and not a clearance opinion** — infringement turns on claim construction, doctrine of equivalents, prosecution history, and validity, none assessed here. Engage patent counsel before relying on any conclusion. Claim texts verified directly from USPTO grant PDFs. **Companion:** [AIQG-PROVISIONAL-BRIEF.md](#)

Techniques assessed

- **(a)** Content-propagation graph (Rabin chunking, TTL index, IDF-weighted, zero-header inference)
- **(b)** `tool_call_id` echo (deterministic flow linkage)
- **(c)** Prefix chaining (hash multi-turn prefix → conversation id)
- **(d)** 8-tier identity-confidence ladder + asserted-vs-inferred impersonation signal
- **(e)** Agent-type fingerprinting (13-signal ensemble, MinHash lineage)
- **(f)** Per-agent / per-flow cost attribution & metering at the gateway

The four patents (verified identities)

Patent	Title	Assignee	Priority	Grant	Independent claims
US 12,483,411 B1	Blockchain-based AI agent life-cycle management & authentication	Tyntre LLC	2025-03-04	2025-11-25	1 (system), 11 (method)
US 12,547,677 B1	Anticipatory (JIT) authentication for account-specific AI agents	Airia LLC	2025-05-16	2026-02-10	1 (method), 19 (CRM), 20 (system)
US 10,924,326 B2	Clustered real-time correlation of trace-data fragments (distributed transactions)	Dynatrace LLC	2015-09-14	2021-02-16	1 (method), 15 (system)
US 11,785,115 B2	Request tracing	IBM Corp	2021-09-27	2023-10-10	1 (system), 11 (method), 18 (CRM)

What each covers (one line): - **Tyntre** — blockchain-stored agent identity token + permitted cross-session context-passing, identity token attached to each response. - **Airia** — pause/resume agent execution to acquire **per-user** (not global) OAuth-style tool credentials via a credentials service. - **Dynatrace** — APM tracing where each agent **stamps an explicit correlation-server ID + transaction ID into outgoing messages**; dependents add per-customer metric attribution + trace splitting at client boundaries. - **IBM** — attribute spans

to a specific process by correlating **socket/port connection data at a sidecar proxy** with span metadata, to disambiguate name-collisions.

Risk matrix (patent × technique)

HIGH = independent-claim language plausibly reads on the technique; MEDIUM = conceptual overlap but a claimed element likely missing; LOW = distant; NONE = no meaningful read.

Technique	Tyntre 12,483,411	Airia 12,547,677	Dynatrace 10,924,326	IBM 11,785,115
(a) Content-propagation graph	LOW	NONE	LOW	LOW
(b) tool_call_id echo	MEDIUM	LOW	MEDIUM	LOW
(c) Prefix chaining	LOW	NONE	LOW	LOW
(d) Identity ladder + impersonation	MEDIUM	LOW	LOW	NONE
(e) Agent-type fingerprinting	LOW	NONE	LOW	MEDIUM
(f) Per-agent metering	LOW	LOW	MEDIUM	NONE

No HIGH-risk read on any technique as the design is described.

Key rationales (tied to claim language)

- **(a) is clear of all four.** Dynatrace and IBM both require an explicit propagated identifier or socket-connection observation; the zero-header content-fingerprint mechanism is the opposite. Tyntre requires content on a blockchain with an attached token. No claim recites content-similarity matching.
- **(b) vs Dynatrace (MEDIUM)** — Dynatrace claim 1 covers an agent *propagating* a transaction id into a message and a downstream agent *retrieving* it. A tool_call_id reappearing is functionally id-in-payload linkage. **Gap that saves us:** Dynatrace requires an instrumented agent that *deliberately appends* the id plus a correlation-server *selection/routing* scheme; the gateway here passively observes an id the model echoed and does no server selection.
- **(b)/(d) vs Tyntre (MEDIUM)** — same problem space (agent identity, carrying context across sessions) but every Tyntre independent claim **requires a blockchain and an attached agent identity token**, which the design uses neither of.
- **(e) vs IBM (MEDIUM)** — IBM disambiguates same-labeled services by *characteristics*; same goal as agent-type fingerprinting. **Gap:** IBM requires those characteristics from **socket-connection correlation at a network traffic component (sidecar proxy)**; the 13 signals here are L7-semantic at an API gateway.
- **(f) vs Dynatrace (MEDIUM)** — Dynatrace dependents 3/4/17 attribute per-transaction metrics per **customer** via appended customer id + trace splitting. **Gap:** the design attributes cost to *agent/flow* via *inference* from the gateway ledger, not to a *customer* via a *propagated* correlation id.

Overall posture

Patent	Posture
Tyntre 12,483,411	Low-to-moderate — clear if no blockchain + no identity token attached to responses.
Airia 12,547,677	Low — out of scope (JIT per-user tool auth).
Dynatrace 10,924,326	Moderate — the one to take most seriously. Broad, 2015 priority, sophisticated assignee, expires ~2035.
IBM 11,785,115	Low-to-moderate — clear if no socket/port correlation at a sidecar proxy.

Clear: (a) content-propagation graph, (c) prefix chaining. **Lightly encumbered, defensible on a missing element:** (b), (d), (e), (f).

Design-arounds for the MEDIUM risks

vs Dynatrace (priority target): 1. Keep flow linkage **inference-only** — never inject a TAS-generated correlation id into outgoing requests; rely on the model-produced `tool_call_id` echo and content propagation. 2. Do **not** implement a correlation-server *listing + selection/routing* scheme. 3. For (f), bill from the gateway’s **own ledger keyed on inferred identity** — not a customer id appended to messages — and avoid “trace splitting at client-control boundaries” as a mechanism.

vs IBM (for (e)): 4. Keep fingerprinting at **L7/application semantics** (toolset hash, prompt template, JSON schema, call-graph, timing); **never** correlate raw socket/port to process at a sidecar proxy.

vs Tyntre (for (b)/(d)): 5. Do **not** store agent identity tokens / permissions / context on a blockchain, and do **not** attach an agent identity token to served responses. Frame impersonation detection as statistical mismatch, not token verification against a ledger.

vs Airia: 6. No change needed; just don’t add a pause/resume-on-missing-credential JIT per-user tool-auth flow to the gateway.

Recommendation

Commission a **counsel-led opinion focused on Dynatrace 10,924,326** and on techniques (b) and (f) **before any billing-grade per-agent metering ships**. The other three patents are narrower (blockchain identity, JIT auth, socket-proxy tracing) and the design differs on a load-bearing element of each. Observability-only use (no billing) carries materially lower exposure.